THE PROBLEM SOLVING ASSOCIATION CLG

IN JOINT ENDEAVOUR WITH

Trinity Floating Wind Team

Theoretical Physics Student Association of Ireland

TPSA HACKATHON 2026 · INFORMATION SHEET

Efficiency in Ireland's Electrical Grid

An investigation into losses, constraint and capacity across the national transmission and distribution network

DOCUMENT

Problem Brief & Information

DATE

September 2026

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AT A GLANCE

Problem Summary

In 2025, 12% of our renewable energy was dispatched down, meaning that wind and solar generation was turned off even when the wind was blowing and the sun was shining. There are three reasons for dispatch-down: curtailment, constraint and surplus. Curtailment refers to dispatch-down of generation for system-wide reasons (e.g. to maintain 50Hz on the grid). Constraint refers to dispatch-down due to congestion on transmission infrastructure. Surplus refers plainly to the scenario where more power is generated than is demanded, and so is refused access to the grid. Around 5–6% of our renewable energy is constrained, and that figure is predicted to rise in coming years.

12–14%
of wind energy wasted due to dispatch-down per year

13,364 GWh
of wind generated across the island of Ireland in 2025

€100M+
of potential savings from optimising storage and dispatch-down

14
teams to tackle the problem

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Organising Committee

Everyone who made the week happen.

COMMITTEE

Seán O'Connor - TF Wind

Artem Tarasenko - TPSA Ireland

Casey Farren-Colloty - TPSA CLG.

SPONSORS

- School of Mathematics
- School of Physics
- School of Engineering
- TCD EngSoc

SPECIAL THANKS TO

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Declan Nell, Harun Šiljak, and Eliza Somerville for speaking.

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1 Introduction

The purpose of this hackathon is to stimulate interesting ideas and research that could contribute societal good. The average Irish household electricity bill is €1,870/year. That results in a total spend of over €3bn every year. On top of that, the taxpayer pays for our energy infrastructure: the CRU has now approved a €18.9bn spend for 2026-2030 to upgrade the grid.

The spend is large and the number of people who understand the system is small. For anyone who is comfortable with complex problems and is a quick learner, the problem area is ripe with opportunities to truly help the average person. Even finding efficiencies to reduce constraint by 1-2% could save millions of euro of wasted energy every year.

For this reason, this year's edition of the hackathon is focused on the electrical grid. While specifically focused on constraint, you are welcome to hone in on any aspect of efficiency in the Irish grid in your team's solution.

This document provides sections on the context of the problem, as well as explanation on concepts which may be useful to you. There is also a github repository available here:

https://github.com/farrencc/Hackathons

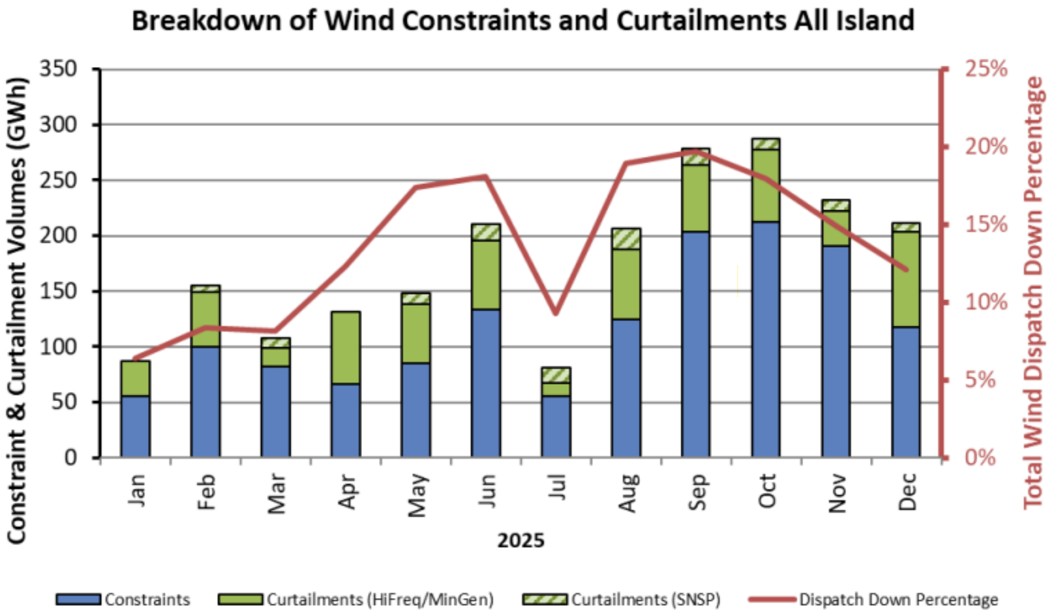
There is no requirement to understand everything in this document, and certainly not to read everything in the github. Your main resource in this hackathon is yourself, your computer, and the teammates around you. Use the documents, supports, and facilities as you see fit to create the most interesting, exciting or impactful solution to the problem of how our electrical grid could be more efficient.

2 Problem Context

There are three reasons for dispatch-down, or having renewable power "turned off":

- **Curtailment** refers to dispatch down of generation for system-wide reasons (e.g. to maintain "inertia" on the grid).
- **Surplus** is simply when demand does not meet the supply of renewables and there is no energy storage which can bridge this time gap
- **Constraint** refers to dispatch down due to congestion on transmission infrastructure. Around 5-6% of our renewable energy is constrained, and that figure is predicted to rise in coming years.

For more detail, it is recommended you read this 2025 EirGrid report on dispatch-down.



Constraint Groups

Constraint groups are how EirGrid turns a network problem into a dispatch instruction. When wind/solar output would overload a line or cause voltage issues under intact, N-1, or N-1-1 conditions, the control room's only real-time lever is to group the potentially responsible farms together and issue them a joint MW cut. Membership is about who can actually relieve the violation and not who's nearby.

Algorithm 0 shows how that membership gets decided: for each unresolved violation, perturb one farm's output at each candidate node by 10 MW, balance it against a fixed remote generator, and measure the resulting change in flow on the monitored circuit. This is defined to be the node's shift factor. Nodes that clear a materiality threshold get pulled in and the rest don't, no matter how close they sit.

We get an output that is a static, predefined group the WDT can fire instantly. These predefined groups can be seen in Figure 2. The whole point of doing the effectiveness study offline is so nobody has to run it live during an event. Although sometimes it is necessary to make groups up on the fly. Splitting the actual MW reduction across group members once it's invoked is a separate step (Appendix 1's pro-rata calc) and isn't part of this algorithm.

```
Input: Base case 𝒞 (low-load EMS case, summer ratings, all renewables at max output, EWIC
       at high export); set of candidate nodes 𝒩 with Category-2 renewable generation;
       contingency set Θ = {base case, N-1, N-1-1}; effectiveness threshold τ
Output: Constraint group 𝒢 ⊆ 𝒩
Apply SEM-11-062 mitigations to 𝒞: dispatch down conventional/lower-priority generation,
  sectionalise transmission equipment where possible
Run base case and all θ ∈ Θ on 𝒞
𝒱 ← set of unresolved thermal/voltage violations after mitigation
foreach violation v ∈ 𝒱 do
    𝒢_v ← ∅
    Identify monitored circuit/element e_v and its binding contingency θ_v ∈ Θ
    Record pre-perturbation flow I_1 ← flow(e_v | θ_v)
    foreach node n ∈ 𝒩 feeding into the region of v do
        // one representative farm per node is sufficient; co-located farms at
           an unsectionalised station share effectiveness
        Select one wind/solar farm f_n at node n
        Reduce output of f_n by ΔP = 10 MW
        Balance +ΔP at a fixed remote conventional generator, far from v
        Re-run case under θ_v; record post-perturbation flow I_2 ← flow(e_v | θ_v)
        SF(n) ← (I_2 − I_1) / ΔP                          // shift factor / effectiveness
        Restore f_n to maximum output
        if |SF(n)| ≥ τ then
            𝒢_v ← 𝒢_v ∪ {n}
        end
    end
    foreach n ∈ 𝒢_v do
        Add all Category-2 wind/solar farms at n (directly connected or via local distribution)
          to 𝒢
    end
end
return 𝒢
```

Algorithm 0: WDT Constraint Group Generation

WIND DISPATCH TOOL CONSTRAINT GROUP OVERVIEW

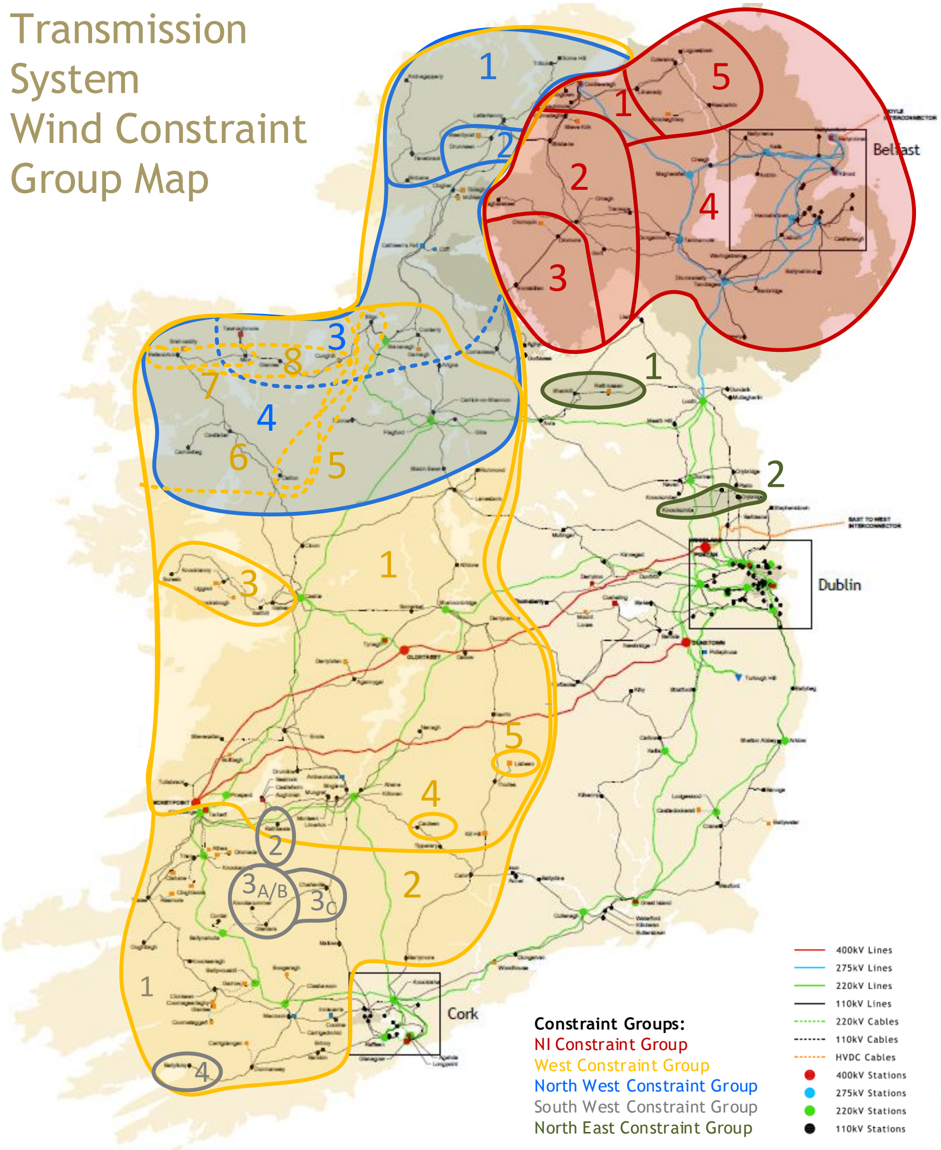


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Figure 2. A map of Irish constraint groups as used in the Wind Dispatch Tool in 2024.

Curtailment

In the Irish power system, curtailment and inertia are linked through the System Non-Synchronous Penetration (SNSP) limit, which caps the share of demand that can be met by non-synchronous generation (wind and solar) at any moment to preserve enough rotational inertia on the grid. Inertia, physically stored as kinetic energy in the spinning mass of synchronous generators, resists sudden changes in system frequency and gives operators time to respond to a fault or a large generator trip-off. As more wind and solar displace conventional synchronous plant, system inertia falls and the rate of change of frequency (RoCoF) that any single credible event can produce rises toward operational limits. When the SNSP limit is reached, EirGrid's Wind Dispatch Tool curtails wind output to bring non-synchronous penetration back within the secure envelope. Critically, this is applied as a single system-wide scalar across the wind fleet, not through the topology-specific constraint groups used for thermal or voltage limits, so curtailment is modelled as one global SNSP/inertia constraint rather than as a zonal or line-specific one.

Surplus

Surplus dispatch down happens when wind generation exceeds demand net of the minimum output that conventional generators must run at to stay online for reserve and stability purposes. Because thermal units can't simply switch off and on to track wind i.e. they have minimum load levels, minimum up/down times, and start-up costs etc. a must-run fleet stays on the system even when wind alone could meet demand, leaving any wind output above that residual headroom with nowhere to go. The result scales with how far wind exceeds must-run demand, and should improve with more flexible thermal plant, interconnection, storage, or demand-side flexibility that can absorb the excess without needing conventional units to idle at minimum load.

3 Problems

The difficulty of these problems will vary quite wildly. Both with realm of the problem, the nature of solving it, and the depth of complexity associated with it. Do not let this affect your choices *too* much. The variance of difficulty found here is the reason why this is the first hackathon in 3 years to not have an associated difficulty level assigned to the different problems. Take the leap and try even if it seems difficult. You never know what you might stumble across.

Constraint

3.1 Analyse the North-West

The map in Figure 3 shows a section of the 110 kV transmission network in the north-west of Ireland, corresponding to EirGrid's NorthWest Constraint Groups **1**, **2**, and **3**.

Here are some research questions to look into on your excerpt of the grid:

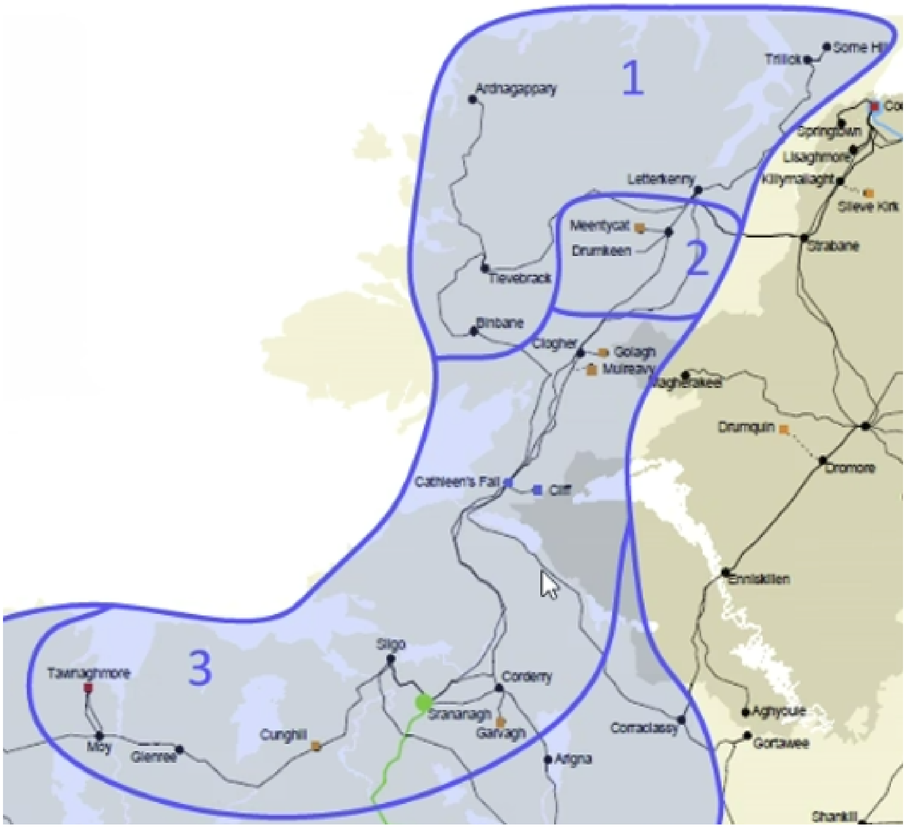

Figure 3. A map showing EirGrid Constraint Groups 1, 2 and 3 and the substations within each group

- Which lines are constrained most often? How does thermal rating affect the amount constrained?
- Propose a battery siting and sizing strategy: which node(s) should host storage, and what power/energy rating would be required to eliminate constraint on your identified line across the full hourly time series provided?
- How could Dynamic Line Rating reduce the amount of wind constraint on your model?
- Could your model be used to indicate to wind/BESS developers where they should locate their plant?
- Can you use this model to calculate the shift factor of each wind farm if it acts through its nearest substation?
- In reality, some wind farms hold priority and non-priority status. Can you factor this in to your model and locate where this causes inefficiencies?

3.2 Constraint Group Generation

Build a correlation map Complex systems as they exist on graph networks, as in the continuous limit, often have bizarre or obscure non-local effects. That is to say, high correlations in quantities measured at geographic distances. It is the goal of this problem to investigate and potentially build a map which shows these non-local effects in the grid. Currently one metric EirGrid uses in this domain is the *shift factor*. It is worth questioning if this is the best tool for the job. [1]

[1] There may be room to apply `PyPSA` to this problem.

Investigate methods of generating groups There are multiple ways that constraint groups can be generated in principle. One of them also uses this shift factor as a metric. One might apply clustering or community detection methods as another means of generation. But these are far from exhaustive. It is important to develop these with the mindset of N, $N-1$, $N-1-1$ fault tolerance that EirGrid abides by.

One current method of generation includes assigning each overloaded line, found via simulation, a separate constraint group and then follow an algorithm of editing these groups. This too is worth questioning i.e. does there need to exist one group for each problematic line.

3.3 Dispatch-down Strategy

Constraint groups are used to dispatch down a set of generation nodes at once. Is there an optimal strategy to ensure a safe and effective incentive structure is enforced?

3.4 Full Grid Analysis

Extend your analysis of the North-West region to cover the entire grid. Although a more computational intensive problem there are other features to consider as well such as interconnectors.

Curtailment

3.5 Local effects

Generally, inertia - and hence curtailment - is seen as a global phenomenon. Indeed, curtailment is usually defined as such. However, there must necessarily be some local effects. The aim of this problem is to analyse and quantify them.

This is a non-trivial task. A potential test may take the form of changing inertia generating energy sources such as gas with an equivalent source of wind or solar energy which do not generate inertia. This, again, should not be considered an exhaustive list.

3.6 Synchronous Condenser Siting

Synchronous Condensers (SC's) are one method of ensuring the grid has the required inertial properties at any given time to safely respond to small fluctuations in demand. The siting of these becomes important in the context of local effects. If one finds non-local effects then the logical next question is where might one place SC's to take advantage of this?

Surplus

3.7 Battery Siting

One may hypothesise that battery siting at or near generation nodes provides an optimal placement by which to reduce surplus dispatch down. This is clearly a very intuitive strategy to pursue but does your analysis corroborate it?

Miscellaneous

These are largely undefined problems or ideas that may be worth looking at but which do not fit the above categories.

3.8 Resilience

Capacity to hold inertia under high stress

One potential scenario in which inertia becomes a local problem is as follows. Suppose the grid network takes a shape such that it can be cleanly split into two subgraphs by cutting one edge (transmission line). If the grid was set up such that one subgraph only had access to wind generation then a relatively local area would severely lack the responsiveness that inertia yields.

Of course that's a relatively contrived example but under sufficient stress the mobility of electrons between clusters may also be called into question. In this case, is there a risk of a seeing a reduction in the effects of inertia?

Artery identification This problem would most likely require access to the distribution network. An unofficial map will be provided to participants which was prepared using OpenStreetMaps. The identification of arteries within the transmission and distribution networks is tantamount to finding high-risk areas. Sections of the network where battery placement and safe-guarding are highly important e.g. areas connected to hospitals. The long-term question there becomes how does one lower the risk associated with these areas?

3.9 Uses of energy

One problem facing the Irish grid is that generation is somewhat restricted by demand. High generation when needed must be financially viable when not. One potentially way of aiding this is to use energy in creative ways. Here are some ideas to investigate but the participants should think of others:

- District heating / cooling
- Interconnectors
- Heavy industry located in strategic places

3.10 New forms of generation

As mentioned there are two fundamental types of energy generation: that which does and does not result in inertia. Two potentially interesting forms of energy generation that would be both clean and produce inertia (which is a significant downside to wind and solar) are nuclear fission - specifically in the form of Small Modular Reactors (SMR's) - and grid-forming batteries. The goal of this problem is to consider what an optimal placement strategy might be for technologies like this. Again, your analysis may not be limited to just considering these.

EirGrid is currently en-route to developing sizable offshore wind capacity. One key difference between offshore and onshore wind generation is how they connect to the grid. Onshore wind farms can be spread throughout a large area with the load being similarly spread out throughout the network while offshore connects to a small number of nodes meaning a high density of input. This difference strikes at a notable difference in strategy and hence one might consider the effects of this change on the grid's behaviour.

3.11 Infrastructure placement to aid in electric vehicle adoption

There is an increasing amount of Battery Electric Vehicle (BEV) registration in Ireland.[2] One issue adoption faces is the chicken-and-egg nature of BEV infrastructure development evidenced by the fact that although Cork and Dublin together make up $\sim 40\%$ of the Republic's population they account for $\sim 50\%$ of BEV ownership. One major question then becomes how and where might we invest in supporting infrastructure to aid in more wide spread adoption. This might take the form of battery siting or distribution network adjustment. The aim of this problem is to come up with recommendations - if any are needed - to assist in Ireland's Electric Vehicle Policy Pathway.

§ References

[1] TPSA Hackathon. Efficiency in ireland's electrical grid: data and scripts. https://github.com/farrencc/Hackathons, 2026.

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[3] Maeve McTaggart. €450m worth of wind energy 'wasted' last year due to pressures on the grid. *Irish Independent*, March 2026. URL https://www.independent.ie/irish-news/450m-worth-of-wind-energy-wasted-last-year-due-to-pressures-on-the-grid/a/138990719.html.

[4] National Energy System Operator. What is inertia?, 2026. URL https://www.neso.energy/energy-101/electricity-explained/how-do-we-balance-grid/what-inertia.

[5] EirGrid. Annual renewable constraint and curtailment report 2025. Technical report, EirGrid, 2025. URL https://cms.eirgrid.ie/publications/annual-renewable-constraint-and-curtailment-report-2025.

[6] Central Statistics Office. Composition of ireland's vehicle fleet 2026. Technical report, Central Statistics Office, July 2026. URL https://www.cso.ie/en/releasesandpublications/ep/p-civf/compositionofirelandsvehiclefleet2026/keyfindings/.

[7] Electric Vehicle Policy Pathway Working Group. Electric vehicle policy pathway: Working group report. Technical report, Government of Ireland, 2021. URL https://assets.gov.ie/static/documents/electric-vehicle-policy-pathway.pdf.

[2] See here for more.

A Data Sources & Reproducibility

The network figure is built from published operator and open data; the spectra are illustrative, generated from a stylised model of the transmission system. Code and data live in the shared repository below; each figure is reproducible from a single script [1].

Artefact	Source	Licence
Transmission network (≥110 kV)	EirGrid Transmission Development Plan 2024 web map	Operator data
Distribution network (<110 kV)	OpenStreetMap (Geofabrik Ireland extract)	ODbL
County boundaries	Ordnance Survey Ireland	CC BY 4.0
System demand, wind & curtailment	EirGrid / SEMO system data	Public sector
Analysis code	github.com/farrencc/Hackathons	MIT

Table 1. Data sources behind the figures and tables in this document.

B GitHub Repository

What you're getting

EirGrid publishes the Ten Year Transmission Forecast Statement (TYTFS) as four PSS/E load-flow cases. We've turned them into PyPSA networks, put coordinates on the buses, extracted a 15-node North-West region, attached a week of hourly profiles, and packaged all of it as a standalone kit with six worked examples. This sheet explains how that data was built, what it can and can't support, and what's in the repository you've been given. Everything here is reproducible from the code; if you want the full derivation behind any number, the long-form report and phase documents (Section) have it.

Headline facts. Four TYTFS cases → eight PyPSA networks (transmission-only and full-voltage scope). All connected, all solve a DC power flow, all solve a linear optimal power flow. The conversion is checked against each case's own solved voltage angles (correlation ≥ 0.99 on seven of eight networks). 87% of transmission buses are geocoded, cross-checked against EirGrid's station register at a median separation of 14 metres. 215 automated tests pass.

Case	**Condition**	**Peak demand**	**Connected capacity**
WP2024	Winter peak, 2024 network	7,325 MW	22,650 MW
SV2024	Summer valley, 2024 network	4,948 MW	14,591 MW
WP2033	Winter peak, 2033 network	8,792 MW	42,574 MW
SV2033	Summer valley, 2033 network	6,058 MW	18,283 MW

WP2033 is the one with the headroom: 42.6 GW of connected plant against an 8.8 GW peak is where constraint and curtailment live. One geocoding wrinkle worth knowing up front: the 2033 cases place only 78% of buses, not 87% the difference is trivial, stations that don't exist yet aren't in OpenStreetMap.

Startup

The participant kit is self-contained and imports nothing from the build pipeline that produced it.

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python examples/a_dc_power_flow.py
```

That's the whole quickstart: a clean environment to a plotted map in well under a minute.

Directory	**Contents**
networks/	4 scenarios × 2 scopes, as .nc and as CSV folders
gridkit.py	load, edit, reset, save a network; read results back
flowmath.py	PTDF, edge-to-edge susceptibility, shift factors
plotstyle.py	one shared palette for every chart
examples/	six worked scripts, a to f
test_kit.py	20 checks, runnable without pytest

How the network was built

A PSS/E raw case is a snapshot of solved voltages, solved dispatch, no coordinates, no time axis. Six decisions turn it into something a capacity-expansion model can use, and each is the kind of thing that produces a network which still seems okay if you get it wrong.

Decision	What we did, and why
Per-unit base	Every case declares SBASE = 100 MVA. We read it and refuse any case on a different base, rather than assume 100 else 200 comes up and one gets a silent factor-of-two error otherwise.
Branch rating	PSS/E carries up to 12 rating slots per branch; only RATE1–RATE3 are populated. RATE1 == RATE2 exactly, in all 4,841 branch records across the four cases, and RATE3/RATE1 is 1.0 or 1.1. We read this as normal/LTE/STE and take RATE1 as s_nom. This is an inference from the data since Siemens' format manual isn't public i.e. it's not a documented property of the format.
Unrated branches	18–20 branches per case carry no rating (RATE1 of 0 or 9999). All are couplers or stubs (busbar sections, capacitor ties, generator terminal ties), not circuits with a forgotten rating. Each is bounded by the smaller of its two ends' summed incident capacity.
Sub-110 kV scope	The kit ships two scopes, transmission-only (110 kV+) and full-voltage. Removing a low-voltage bus would strand its generation, so it's aggregated into the retained bus it reaches by the least-reactance path instead of simply deleted.
DC interconnectors	Moyle, EWIC and Greenlink become PyPSA Link components (controllable injections), not part of the linear AC network, since the file doesn't model the far (GB) side of two of them.
Reference bus	The case names its own swing buses (PSS/E bus-type code IDE = 3). Getting this wrong routes the whole network's power imbalance through whatever generator PyPSA picks by default. In one case, 856 MW through a 21 MW hydro unit. Fixed by assigning the reference as a full column after the network topology is determined.

Each of the eight networks is checked against the case's own solved voltage angles and not against itself. If a reactance, a transformer equivalent or a phase-shift sign were wrong, a linear power flow's angles would not track the file's. Angle correlation is ≥ 0.99 on seven of eight networks (the eighth, WP2033 full-voltage, is 0.90 i.e. more buses, more chances for a small disagreement to compound).

Coordinates and the North-West region

Bus records carry no coordinates and the only join key is a twelve-character station name. Every named power=substation and power=plant on the island was pulled from OpenStreetMap (1,705 named objects, 367 at 110 kV or above) and matched against station names. Nothing is interpolated or guessed: a station that can't be matched is left uncoordinated rather than placed by a heuristic. 87% of transmission buses were matched, cross-checked against EirGrid's own Transmission Development

Plan station layer at a median separation of 14 metres.

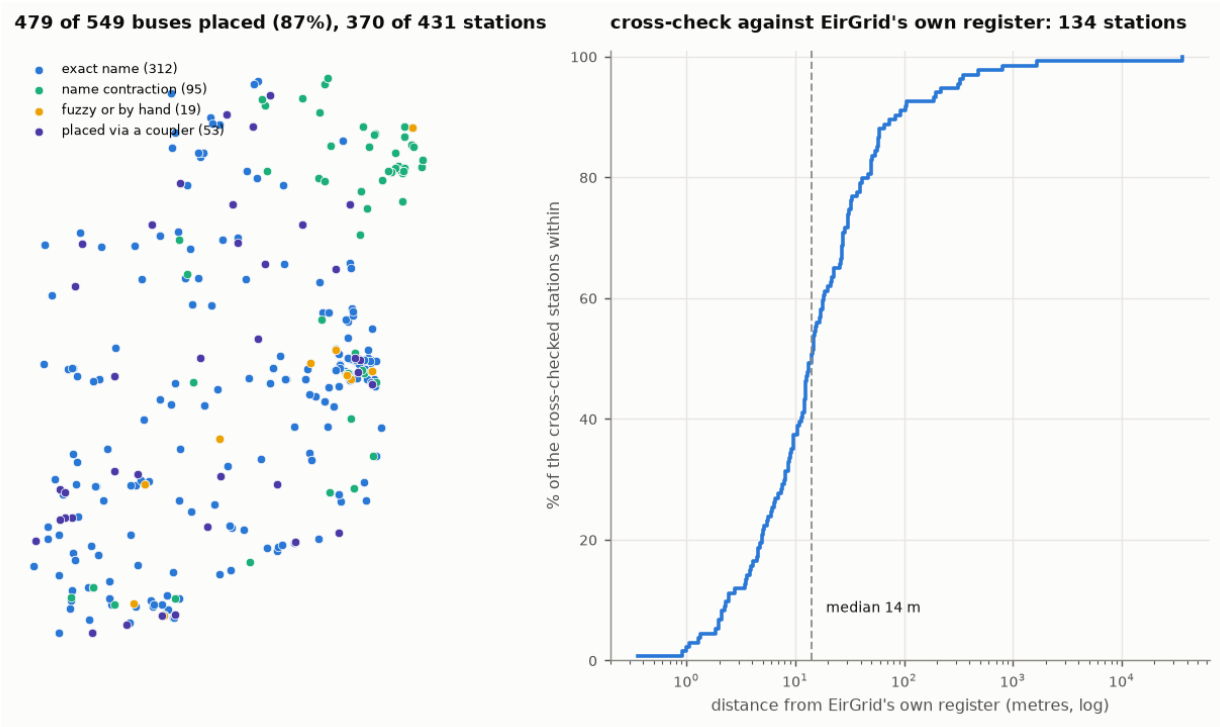


Figure 4. Left: how each WP2024 transmission bus was matched. Right: the cross-check against EirGrid's own station register with a median disagreement 14 m.

The North-West region (Donegal/Sligo/Mayo) is extracted as a 15-node subnetwork and reconciled against a hand-built dataset. Two things worth knowing before you build on it: both north-western cross-border ties (to Strabane and to Enniskillen) run through a phase-shifting transformer whose tap this model holds fixed so a tie's flow limit here is a thermal one, not a control-action one. And line lengths in the TYTFS files are real, not placeholders: 83% of transmission branches carry a length, and it correlates at $r = 0.983$ with the great-circle distance between independently geocoded endpoints though seven real circuits (six of them in Northern Ireland) carry no length at all, and a per-km calculation should skip those rather than treat the gap as a real zero.

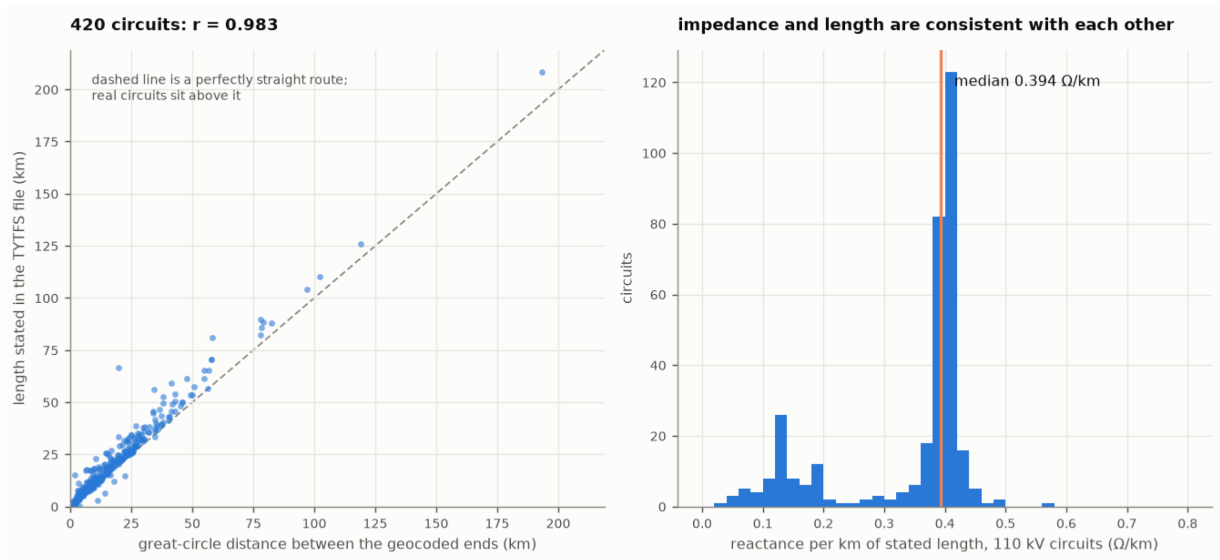


Figure 5. Left: stated length vs. great-circle distance between geocoded endpoints, 420 circuits. Right: reactance per km separates cleanly into overhead line and cable.

The 15-node view has one costly fold. Merging Srananagh's two busbars into a single node short-circuits the 250 MVA transformer between them, introducing up to 21.6 MW of flow error against the full network. Almost all of the view's total error comes from this one fold. Splitting Srananagh back into its 110 kV and 220 kV buses (16 nodes instead of 15) cuts that to 1.7 MW. If your work is sensitive to flow on that transformer, make this split before trusting the numbers.

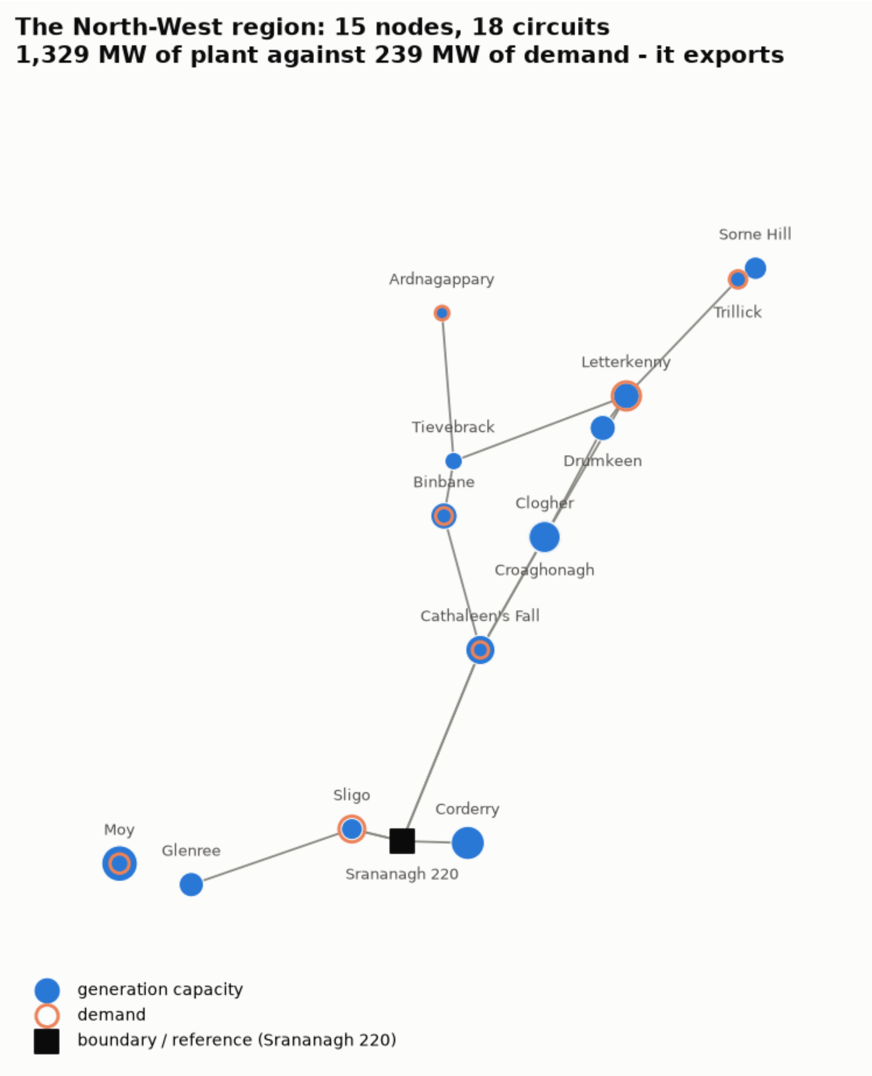


Figure 6. The region as the kit ships it. Dot area is plant capacity, the ring is demand, the square is Srananagh 220 kV with the boundary and the reference.

The hand-built dataset has a units bug worth knowing about. Cathaleen's Fall, Cliff and Golagh hydro plants all show capacity "38" in the hand-built data this, like others, is not a real MW figure (Cathaleen's Fall is actually two machines of 22.5 and 23 MW in the TYTFS file). Separately, the 24-node and 15-node views of this data disagree by exactly Cunghill's 35 MW, because one view includes it and the other doesn't.

Time series data is synthetic

TYTFS gives four snapshots. A constraint or curtailment study needs hours, and a real weather pipeline was not used. Instead, synthetic.py generates a full year of profiles as a spatially correlated random field: an exponential correlation decay with distance ($L = 400$ km) and an Ornstein–Uhlenbeck process in time, mapped through a Gaussian copula to a Weibull wind distribution. Independent per-site noise would have made dispatch-down behaviour meaningless. 400 sites of uncorrelated noise barely leave the fleet's mean output, and the network is never asked to carry anything. The correlation actually produced by the generator was measured back off its own output and checked against the field it was built from; they agree.

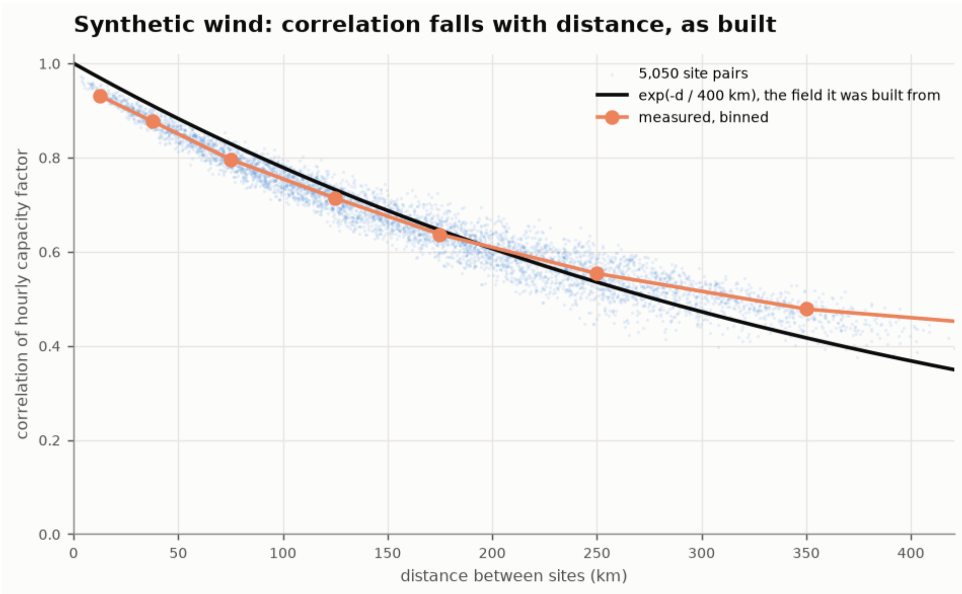


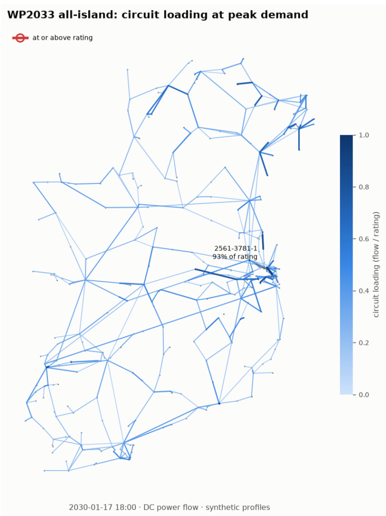
Figure 7. 5,050 site pairs: the correlation the generator actually produced (orange) against the exponential decay it was designed to follow (black).

Each scenario's week of profiles is anchored on an hour of the generated year chosen to reproduce that case's own state (winter peak or summer valley), tapering back to the unconstrained field over 18 hours either side. The generator is seeded (SEED = 42), so it's deterministic and reproducible.

The participant kit

Script	Demonstrates	Notes
(a)	DC power flow	Takes a given dispatch and reports where the flow ends up; nothing here is optimised.
(b)	Dispatch-down split	Least-cost LOPF over the week, split into constraint-based and surplus-based dispatch-down. Not the same thing as SEM/EirGrid curtailment, which is a system-wide SNSP mechanism this model doesn't represent.
(c)	Capacity expansion	Marks binding circuits extendable, offers a battery at every renewable bus, solves.
(d)	PTDF	Built from the pseudoinverse of the weighted graph Laplacian; checked against a power flow to roughly 10^{-6} MW.
(e)	Edge-to-edge susceptibility	How reinforcing one circuit shifts flow on every other.
(f)	Shift factors	The Wind Dispatch Tool calculation, reproduced from the same linear algebra as (d).

Sample output from each script (all WP2033; (a), (b), (d), (e), (f) all-island, (c) north-west). Note on (b) and (c): both the plot titles and the underlying gridkit.curtailment() function use "curtailed"/"curtailment" throughout. This is a mistake. Read "curtailed" in these two figures as constraint-based + surplus-based dispatch-down, not SEM/EirGrid curtailment.



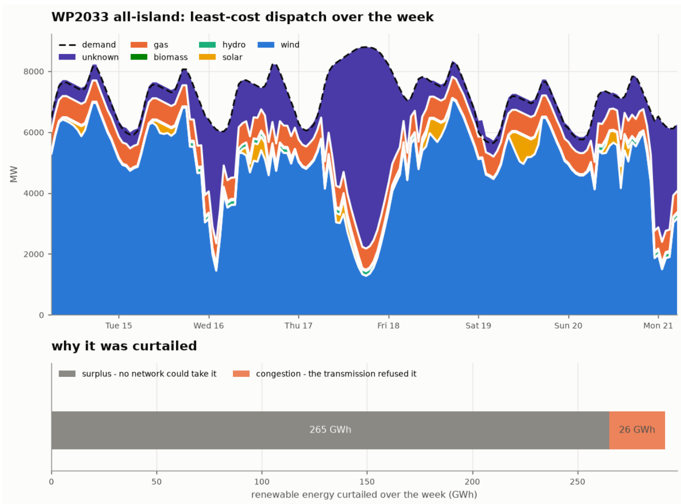


Figure 8. What each of the six example scripts produces, (a) top-left to (f) bottom-right.

Potential issues

freeze_dispatch. n.optimize() writes its answer to generators_t.p. n.lpf() reads generators_t.p_set instead. Run lpf straight after optimize without freezing dispatch first and you get a power flow with no generation in it: the reference bus supplies the whole island, and a circuit can show at 1600% of its rating. Use gridkit.freeze_dispatch().

Buses at 0°N 0°E. PyPSA's netCDF writer turns a missing coordinate into 0.0 rather than NaN, and 0°N 0°E is in the Gulf of Guinea. Plot raw n.buses and Ireland is squashed into the corner of an Atlantic-to-equator map. Use gridkit.placed_buses(n), which drops those and respects the coordinate_source column. These end up being external interconnectors.

A component with no profile silently runs flat out. PyPSA defaults a missing p_max_pu series to 1.0 rather than erroring. Add your own generator or battery without giving it a profile and it will happily run at full output for every hour, no warning given. This is exactly how 7.8 GW of "wind" once ran flat out during this kit's build.

Folding machines together loses the carrier split. If you merge several generators at one station into a single unit, do it per carrier. Folding wind, solar and everything else into one generic unit (as an early version of the North-West extraction did) makes a wind-dominated region incapable of dispatching-down anything, because there's no "wind" left to lower.

Shift factors depend on a reference you choose. shift_factors() offers three conventions for where the balancing megawatt is assumed to come from: spread over loads (reference="load"), spread evenly over every bus (reference="uniform", what the pseudoinverse gives you statically), or all at one named bus (the textbook slack convention). The absolute numbers differ between them but the ranking doesn't. Use any one for ranking generators against a constraint and treat the raw magnitude as convention-dependent.

Limitations

- **This is a DC approximation of an AC model.** The TYTFS files are full AC load-flow cases with reactive power, voltage magnitudes, tap changers and shunt compensation. All of that is discarded: flows are linear in angle differences, voltages are 1.0 pu everywhere, losses are zero. Standard for transmission constraint work and good to a few percent on real flows but it says nothing about voltage, reactive support or stability.

- **Ratings are TYTFS RATE1, not operational limits**, and the normal/LTE/STE reading of the rating slots is an inference, not a documented property of the PSS/E format. A control room works to seasonal, weather-dependent and dynamic limits not present in these files. Ratings also change between scenarios so the same circuit can carry a different rating in each of the four cases.

- **Every time series is synthetic.** No hour in them ever happened. They are anchored to reproduce each scenario's TYTFS state and their spatial correlation is measured rather than assumed, but they are not a historical year and not a forecast. Generators without a geocoded site borrow a neighbour's profile, making those pairs perfectly correlated when they shouldn't be.

- **Costs are placeholders**, not a real price list: gas 90, biomass 40, hydro 1, imports 150 EUR/MWh; wind and solar bid -1 (the mechanism that makes a support-scheme generator willing to pay to stay on, so being dispatched down is a last resort rather than a free choice); load-shedding at 10,000 EUR/MWh, a round number chosen to sit far above everything else and most certainly *not* the Single Electricity Market's real Value of Lost Load (EUR 12,533/MWh, 2022). Only differences between two solves' objectives are meaningful.

- **About a fifth of generation carries the carrier "unknown".** Carriers are inferred from bus-name prefixes and a keyword list; the files don't label plant by technology. Any result that turns on the split between gas and unknown is unsupported.

- **DC interconnectors are simplified**: fixed efficiency, no ramp limits, no minimum stable export, no market coupling. Interconnector flow here is what the cost assumptions produce, not what a day-ahead market would do.

- **An unmodelled control device sits on the constrained north-western tie.** The model prices export to Northern Ireland at nothing beyond the tie's thermal limit; in reality the tie terminates at a phase-shifting transformer whose tap this model holds fixed, and SONI would redispatch around a binding constraint.

- **The TYTFS data is EirGrid's**, provided for reference purposes only. It describes a planning forecast, not the operational system, and is published without warranty as to accuracy or fitness for any particular purpose. **Nothing in this kit is an EirGrid product or an EirGrid position.**

Where everything is

File / directory	What it is
participant-kit/	Everything in this section i.e. the standalone kit
docs/report/report.pdf	The full report: every decision and measurement
docs/PHASE2_PYPSA.md	Every conversion decision, with its evidence
docs/PHASE3_NORTHWEST.md	The North-West reconciliation, table by table
docs/PHASE4_PROFILES.md	The ERA5 pipeline and why it's blocked here
docs/SYNTHETIC_DATA.md	What's real and what's fabricated, in full detail

WORK WITH US

Let's make every megawatt count.

This event is just a starting point. There are two clear ways to take the work forward with the TPSA CLG, the Theoretical Physics Student Association of Ireland and the Trinity Floating Wind Team.

PARTNER WITH US

Partner with the TPSA CLG on our next hackathon

Set a real problem from your organisation and put Ireland's brightest physics and mathematics students to work on it over a focused week. Sponsorship and problem-partner places are open for the next event.

Get involved: problemsolving@tpsa.ie

TAKE IT FURTHER

Contact us to continue building on this work

The methods here scale from a single region to the whole island system. Talk to us about a pilot, a deeper study, or turning these findings into decisions your teams can act on.

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